

System Testing and Development Process Model

6. System Testing Method and Justification

- Chosen Method: Black-Box Testing

Black-box testing focuses on testing the functionality of the system without needing to know the internal code or structure. Since the Inventory Management System is feature-driven, with actions like “login,” “add product,” “record sale,” and “generate alert”; black-box testing is ideal for verifying that each feature works as expected based on input/output behavior.

- You can test:
 - Login functionality (valid/invalid users)
 - Add product form (required fields, validation)
 - Sale processing (stock deduction, alert trigger)
 - Product listing (correct info shown)
- Why it fits:
 - Users interact via a UI and expect predictable results
 - We're testing functional correctness, not internal code logic
 - Allows testers to work without coding knowledge

7. Software Development Process Model and Justification

- Chosen Model: Iterative Development Model

The Iterative Model involves building the software in small, functional versions (iterations); each one improving or expanding the previous. For the inventory system, this allows for:

- Early feedback from working software
- Easier debugging and problem isolation
- Progressive improvement of features
- Reduced risk of complete failure

- Sample Iteration Plan:

Iteration 1: Login + User Roles

Iteration 2: Product Management

Iteration 3: Sales Entry + Inventory Updates

Iteration 4: Low Stock Alerts + Testing

Iteration 5: Final Touches + Presentation